

ROCK FALLS, IL, USA

WMO: 725326

Lat: 41.743N

Lon: 89.676W

Elev: 647

StdP: 14.36

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04894

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.3	(c) -0.3	(d) -15.5	(e) 2.4	(f) -5.7	(g) -9.0	(h) 3.5	(i) 0.4	(j) 28.5	(k) 22.5	(l) 25.9	(m) 19.8	(n) 6.2	(o) 280	(p) 0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.6	(c) 90.6	(d) 75.4	(e) 88.4	(f) 74.3	(g) 85.9	(h) 72.7	(i) 78.2	(j) 87.7	(k) 76.5	(l) 85.2	(m) 74.7	(n) 82.8	(o) 8.1	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.2	135.5	84.5	73.3	126.7	82.1	72.1	121.8	80.8	42.3	87.8	40.3	84.8	38.6	82.9	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.1	(b) 21.0	(c) 18.6	(e) -13.9	(f) 95.5	(g) 6.2	(h) 3.2	(i) -18.3	(j) 97.7	(k) -21.9	(l) 99.6	(m) -25.4	(n) 101.4	(o) -29.9	(p) 103.7		
(a) 25.1	(b) 21.0	(c) 18.6	(e) -15.4	(f) 81.6	(g) 6.1	(h) 2.7	(i) -19.8	(j) 83.6	(k) -23.4	(l) 85.2	(m) -26.8	(n) 86.7	(o) -31.2	(p) 88.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	21.6	25.7	37.4	49.8	60.8	70.5	73.1	71.2	64.6	52.6	39.2	27.3		
	DBStd	20.38	12.66	12.45	10.74	9.09	8.58	6.50	5.39	5.15	7.84	9.21	10.05	11.43		
	HDD50	3231	881	683	413	110	9	0	0	0	2	80	347	706		
	HDD65	6433	1346	1101	857	461	189	23	3	9	103	398	775	1168		
	CDD50	3090	0	2	22	105	344	616	718	659	439	160	22	3		
	CDD65	816	0	0	1	5	59	190	256	202	90	13	0	0		
	CDH74	7918	0	0	6	106	609	1839	2473	1741	974	168	2	0		
	CDH80	2603	0	0	0	16	178	644	863	537	326	39	0	0		
(14) Wind	WSAvg	7.9	9.1	9.1	9.5	10.2	8.7	6.7	5.6	5.2	6.1	7.6	8.4	8.4		
(15) Precipitation	PrecAvg	37.0	1.7	1.6	2.4	3.6	4.5	4.9	3.9	4.3	3.0	2.7	2.7	1.8		
	PrecMax	48.3	3.0	3.8	5.2	6.0	10.0	11.3	10.0	7.5	7.1	8.1	5.3	3.5		
	PrecMin	23.8	0.4	0.3	0.7	1.6	0.6	1.2	0.4	1.3	0.8	0.4	0.3	0.4		
	PrecStd	6.1	0.7	0.9	1.6	1.4	2.4	2.9	2.0	1.7	1.7	1.9	1.4	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	61.9	73.1	82.4	90.0	92.7	94.4	92.4	91.3	84.3	71.8	60.9		
		MCWB	50.5	54.3	60.1	62.5	71.7	74.7	79.1	76.2	74.4	69.9	59.4	56.5		
	2%	DB	45.9	53.7	64.6	77.9	84.3	90.1	90.4	88.6	88.0	79.1	63.8	53.6		
		MCWB	42.3	48.1	55.3	60.7	67.5	73.9	77.4	75.6	72.4	64.0	54.7	49.4		
	5%	DB	41.1	46.3	59.4	72.3	81.2	87.6	88.0	85.5	83.6	73.1	59.5	46.1		
		MCWB	38.2	41.6	51.9	57.6	65.7	72.7	76.0	73.9	69.9	61.5	52.8	42.8		
	10%	DB	36.9	41.1	54.0	66.3	76.8	83.7	84.3	82.3	80.6	69.0	54.9	42.5		
		MCWB	34.4	37.3	46.8	54.7	63.5	70.7	73.7	71.2	66.9	59.6	48.5	38.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.7	55.0	62.1	66.4	73.6	79.0	81.9	79.0	77.2	71.8	61.0	57.4		
		MCDB	53.8	58.5	71.2	77.5	85.3	88.7	91.1	89.3	88.0	82.1	67.1	61.3		
	2%	WB	43.1	47.6	57.3	62.4	70.8	76.2	79.1	77.1	74.4	67.2	57.3	49.5		
		MCDB	45.8	52.2	63.6	73.4	81.5	85.8	88.2	86.6	83.4	73.9	62.9	52.8		
	5%	WB	38.3	42.1	51.9	59.6	68.1	74.2	77.4	75.2	71.8	63.6	53.3	43.6		
		MCDB	41.1	46.5	59.3	70.0	77.5	83.6	85.9	83.1	80.6	71.5	58.5	46.5		
	10%	WB	34.8	37.6	46.7	56.2	65.7	72.1	75.3	73.2	69.0	60.1	49.4	38.5		
		MCDB	37.1	41.2	53.0	66.0	74.2	81.3	82.6	80.5	77.3	67.4	55.3	41.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	16.3	18.4	21.7	20.8	20.8	20.6	20.4	23.8	21.6	17.4	14.4		
		MCDBR	17.7	21.0	26.3	30.1	25.1	23.7	22.3	23.2	27.2	28.1	24.4	18.5		
	5% WB	MCWBR	15.2	16.7	18.0	17.8	14.1	12.0	12.0	12.7	13.5	16.6	17.8	15.8		
		MCWBR	16.9	20.6	24.7	25.9	22.0	20.8	20.3	20.2	22.7	24.1	21.4	17.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.289	0.304	0.341	0.388	0.421	0.425	0.447	0.425	0.397	0.344	0.303	0.286		
		taud	2.410	2.379	2.377	2.257	2.227	2.262	2.188	2.247	2.297	2.446	2.501	2.439		
	Ebn at Noon	Ebn at Noon	274	287	287	279	272	270	263	265	264	269	267	265		
		Edh at Noon	25	30	34	42	44	43	45	42	37	28	23	23		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	550	857	1175	1478	1725	1948	1932	1708	1434	946	620	455		
	RadStd	RadStd	63	102	103	127	117	154	118	106	122	112	52	52		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a) Station Only	(b) Regional (55 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(a) Station Only	(b) Regional (55 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A

Nomenclature: See separate page